

PSL P- Revision Final

Unit 2

- Central limit thm
- Joint Probability distribution (discrete)
- Joint Probability distribution (continuous)

Unit 3

• Hypothesis Testing

• Non Parametric Testing

Sign test

Wilcoxon test

T test

(small sample)

For 1 sample

For 2 samples

Z test

(large sample)

For 1 sample

For 2 sample

χ^2 test / χ^2 test

Goodness of fit

F test — variance test

Unit 1 (Prob)

- Erlang distribution
- Gamma distribution
- Exponential distribution
- Poisson distribution
- [Q. Prove that Poisson dist is limiting case of Binomial distribution]
- Binomial distribution
- Bayes Theorem
- Mean, Variance, CDF, PDF, standard deviation

Unit 4

(Optimization Technique)

• Formulation of LPP

• Simplex Method

Direct

Big M

Revised Simplex

• Transportation Problem

North West Complex

VAM $\rightarrow$ Modi/UV

• Assignment Problem (Hungarian method)

* Chi square test use only when probability is used (binom dist etc) in question if it is not mentioned

* T test use when small sample is there $n < 30$

* Z test use when large sample is there $n > 30$

Q2 - Joint Probability (Bivariate) distribution question (8 marks)

Q: 3 fair points are tossed let x denote the no. of heads on first two coins & y denotes tails on last two coins.

Find joint probability of x & y .

② Find the conditional probability of y given $x=1$

Ans. $-1/4$

③ Find covariance (x, y)

④ Find $\rho(x, y)$ (rank of correlation / correlation coeff)

A: Sample space = $\{HHH, HTH, HHT, THH, THT, TTH, TTT\}$

$x \backslash y$	$y=0$	$y=1$	$y=2$	$y=3$	Add	$P(x)$
$x=0$	0	$1/8$	$1/8$	0	$2/8$	
$x=1$	$1/8$	$2/8$	$1/8$	0	$4/8$	
$x=2$	$1/8$	$1/8$	0	0	$2/8$	
$x=3$	0	0	0	0	0	
Add	$2/8$	$4/8$	$2/8$	0	1	1

② Conditional probability = $\frac{\text{Marginal probability of } (x, y)}{\text{Probability } (X=a)}$

Marginal Probability	$\begin{aligned} &= P(Y=0 / X=1) = 1/8 \\ &= P(Y=1 / X=1) \\ &= P(Y=2 / X=1) \end{aligned}$	Cond ⁿ	$\left. \begin{aligned} &1/8 / 4/8 = \frac{1}{4} \\ &2/8 / 4/8 = \frac{2}{4} \\ &1/8 / 4/8 = \frac{1}{4} \end{aligned} \right\} Y \text{ has no fixed value}$

$$\textcircled{3} E(XY) - E(X)E(Y) = \text{Covariance}$$

$$E(X) = \sum x P(x)$$

$$= 0 \times \frac{2}{8} + 1 \times \frac{4}{8} + 2 \times \frac{2}{8} = 1$$

$$E(XY) = \sum \sum xy P(x, y) = 3/4 \quad \text{Ans}$$

$$E(Y) = 1 \quad \text{Ans}$$

$$\textcircled{4} \rho(x, y) = \frac{\text{Cov}}{\sigma_x \cdot \sigma_y} \quad \sigma = \text{stand dev.}$$

$$\text{var}(x) = E(x^2) - (E(x))^2$$

$$\downarrow \quad \downarrow$$

$$\sum x^2 P(x) \quad (\text{mean})^2$$

$$\sum x^2 P(x)$$

$$= 0 + 1^2 P(x) + 2^2 P(x)$$

$$= 1 \times \frac{4}{8} + 4 \times \frac{2}{8}$$

$$\text{Stand dev } (\sigma) = \sqrt{\text{Var}(x)}$$

Q2 - Joint Continuous Distribution -

Let the joint PDF $f(x,y) = (x+y)$

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

Joint k liye
double
integration

Find $E(x) = \iint x f(x,y) dx dy$

~~E(x)~~

$$\iint x f(x,y) dx dy = \int dx \int x f(x,y) dy$$

inner $\int$ integ
y ke respect
mein
outer $\int$ integ
ko x ke
respect mein

$$= \int_0^1 dx \int_0^1 x(x+y) dy$$

$$= \int_0^1 x dx \int_0^1 (x+y) dy$$

$$= \int_0^1 x dx \cdot \left[xy + \frac{y^2}{2} \right]_0^1$$

$$= \int_0^1 x dx \cdot \left(x + \frac{1}{2} \right)$$

$$= \int_0^1 \left(x^2 + \frac{x}{2} \right) dx = \left[\frac{x^3}{3} + \frac{x^2}{4} \right]_0^1$$

$$= \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$\text{Rank: } \rho(x, y) = \frac{\text{Cov}}{\sigma_x \sigma_y}$$

$$\text{Cov} = E(xy) - E(x)E(y)$$

U3 - Hypothesis Testing (Do theory also - definition etc)

S1: H_0 : Find Null hypothesis

S2: H_1 : Alternate hypothesis

S3: Calculate using formula of test asked (calculated value)

S4: Conclusion

* Calc value < tabulated value then H_0 is accepted

ONLY for hypothesis tests (Chi, F, Z, T)

* Calc value > tabulated value then H_0 is accepted

ONLY for non parametric tests
(Wilcoxon & Sign)

U3 - Chi square testing (super imp) -

Q: Records taken the no. of male & female birth in 800 families having 4 children each. Test if the data are consistent with hypothesis that binomial distribution holds true for male & female birth i.e. the probability having male & female child are equal.

No. of male birth	0	1	2	3	4
No. of female birth	4	3	2	1	0
No. of family	$N \times P(X=0)$	$N \times P(X=1)$	$N \times P(X=2)$	$N \times P(X=3)$	$N \times P(X=4)$
	32	178	260	236	94

$$p = 1/2$$

$$q = 1/2$$

$$n = 4$$

$$N = 800$$

$$\text{Binomial } P(X=x) = {}^n C_x p^x q^{n-x}$$

$$P(X=0) = {}^4 C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{4-0}$$

$$P(X=1) = {}^4 C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{3-1}$$

⋮

Though it is given in table but just nahi diya to hese karog for expected value table on next page

We have to see now that binomial dist is fit for it or not.

No. of families (O_i) [Observed]	E_i (Using prob) [Expected value]	
32	$N \cdot P(X=0) = 50$	Solve
178	$N \cdot P(X=1) = 200$	it
260	$N \cdot P(X=2) = 300$	poora
236	$N \cdot P(X=3) = 200$	like shown
94	$N \cdot P(X=4) = 50$	on last page

S1 → H_0 : Let binom dist holds true for given data
S2 → H_1 : Doesn't hold true (assumption)

To always write as we are assuming

S3 → $\chi^2_{cal} = \frac{(O_i - E_i)^2}{E_i}$ do for all 5 sets (families & put in table)

$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
		$\vdots$
		Σ summation kardo table ka of this column

Given in question $\chi^2_{tabulated}$ at 5% $LDS = 9.488$
Nikala $\chi^2_{cal} = 54.433$

S4 → We fail to accept H_0 & assumption of H_0 is false